

Welcome to Coding

MYP 1

Think About It!

- How does a game know what to do?
- How does YouTube play a video when you click it?
- How does a robot follow instructions?



What is coding?

Coding is writing instructions that tell a computer what to do.

Examples

- Playing a game
- Using an ATM
- Watching videos online
- Sending messages



What is Programming?

Programming is the process of creating a set of instructions for a computer.

Example

- Making tea:
- Boil water
- Add tea powder
- Add milk
- Serve



Coding is Everywhere

Where do we see coding?

-  Mobile Apps
-  Games
-  Cars
-  Robots
-  ATM Machines
-  Websites



How Coding Connects to Design

Problem



Design Idea



Algorithm



Coding



Interactive Solution

In Design class, we will learn how to identify problems and plan solutions.

In Coding class, we will learn how to build those solutions.



What is Pictoblox?

Pictoblox is a block-based coding platform.

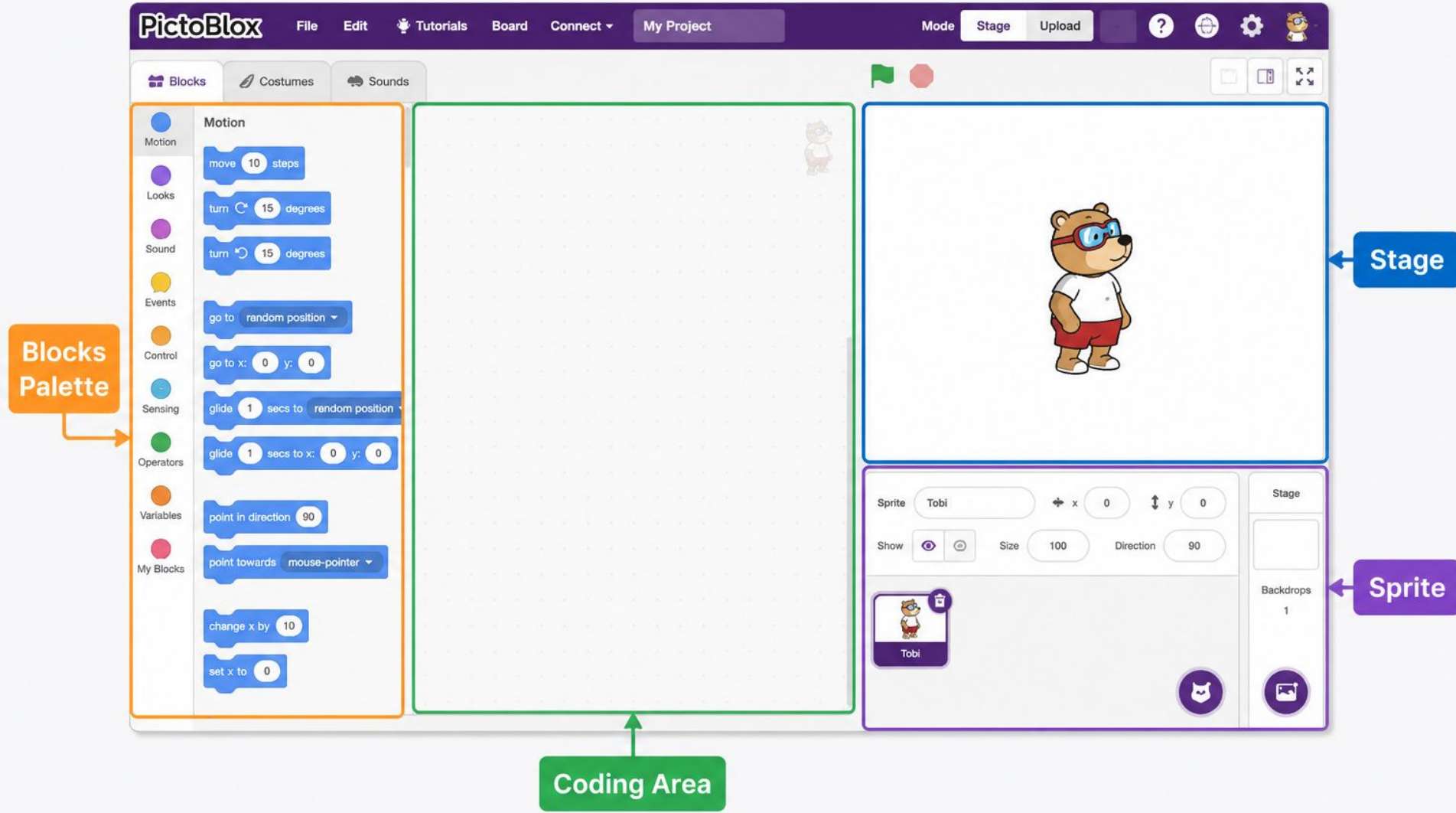
Why use Pictoblox?

- ✓ Easy to learn
- ✓ Drag and drop blocks
- ✓ No need to type difficult code
- ✓ Great for beginners



Pictoblox Interface

Let's get to know the different parts of Pictoblox.



What is sequencing?

Sequencing means arranging instructions in the correct order.

Example

- Getting ready for school:
- Wake up
- Brush teeth
- Wear uniform
- Come to school

If the order changes, the result changes.



Real-Life Sequencing Activity?

Arrange these correctly:

- Eat breakfast
- Wake up
- Reach school
- Wear uniform



First Program

Program Goal

- Make the sprite say:
- "Hello Everyone!"

Blocks:

- When Green Flag Clicked
- Say "Hello Everyone!"



Challenge:

Make the sprite say:

- Your name
- Your favourite food
- Your favourite game

Example:

- "Hi! My name is Aarav."



Input and Output

Input

Information given to the computer

Examples:

- Keyboard
- Mouse
- Touch Screen
- Microphone

Output

Information shown by the computer

Examples:

- Screen
- Speaker
- Printer



Introduction to Variables

A variable is like a box that stores information.

Example:

- Student Name = "Rahul"
- Age = 10
- Favourite Colour = Blue



Thank you



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